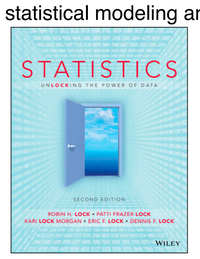
A Catalyst to Understanding Secondary Mathematics Teachers' Knowledge and Transitional Conceptions of Statistical Models

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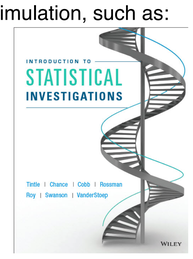
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Introduction

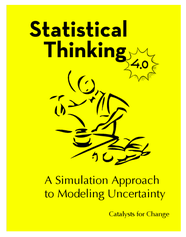
- Recent introductory statistics curricula have been developed that focus on statistical modeling and simulation, such as:



Lock et al., 2017



Tintle et al., 2015



Zieffler et al., 2017

- These curricula have been used at both college and secondary levels.
- Research indicates that these curricula are associated with improved student understanding of statistical concepts and student attitudes towards statistics (e.g., Beckman, delMas, & Garfield, 2017; Chance, Wong, & Tintle, 2016). To date, little of this research has been focused at the teacher-level.
- This study explores the following research questions:
  1. **How do teachers who teach a simulation-based statistics course view statistical models, particularly in contrast to mathematical models?**
  2. **When designing or evaluating statistical models in the context of simulation-based methods, what competencies and misunderstandings do teachers demonstrate**?
- Answers to these questions have implications for the professional development that both pre- and in-service teachers may require to effectively teach simulation-based statistical concepts.

Methods: Participants

- Four secondary mathematics teachers participated in this study, purposefully selected based on their education and varied experiences teaching statistics.
  - Undergraduate degrees: 3 in mathematics education, 1 in mathematics
  - Master's degrees: 3 in education
  - No one had more than two undergraduate courses in statistics
  - 12–25 years overall teaching experience
  - 0–10 years previous statistics teaching experience
    - 2 had taught Advanced Placement (AP) Statistics
    - 1 had taught a course in probability
- Each participant had taught one year of an introductory statistics course as part of the College in the Schools (CIS) program at the University of Minnesota (Zieffler & Huberty, 2015). CIS teachers are experienced school faculty whose qualifications were vetted for teaching the university-level.
- The CIS statistics course used the CATALST curriculum (Zieffler & Catalysts for Change, 2013), based on pedagogical principles founded in research (see Garfield, delMas, & Zieffler, 2012).

References

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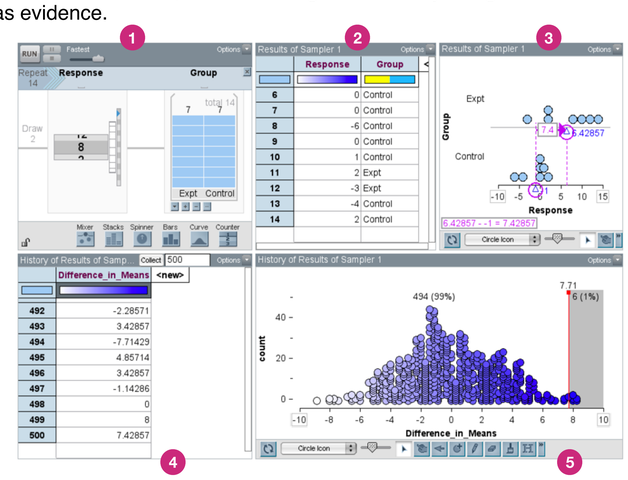
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Methods: TinkerPlots™

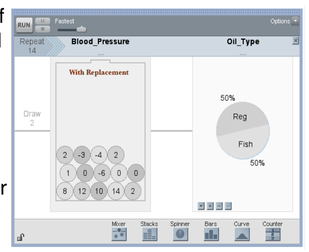
- The general framework through which inference is carried out in CATALST is summarized by ***model–generate–evaluate***. In this framework, students investigate research questions by *modeling* the data generating process using TinkerPlots™ software (Konold & Miller, 2011), *generating* simulated data from the model, and *evaluating* the model by using the observed data as evidence.



  - ***Model* (Pane 1):** The model mimics the data generating process by randomly assigning a group label to each of the response values. Each are sampled *without* replacement, and the group labels are sampled independently of the response values.
  - ***Generate* (Panes 2, 3, and 4):** The model is used to generate one potential random assignment. The response values and randomly assigned group labels for this assignment are shown in Pane 2, and a plot of the response values conditioned on group is shown in Pane 3 along with the mean difference between the two groups. This mean difference is then collected and the process, which is automated, is repeated 500 times (shown in Pane 4).
  - ***Evaluate* (Pane 5):** The 500 mean differences from the simulation are plotted in Pane 5. This distribution shows the variation in mean differences expected under the assumed model. Students then evaluate the model by determining whether the mean difference in the observed data seems compatible with the expected variation. For example, a mean difference of 7.71 (shown with a red line) would not be likely given the model used in Pane 1; we see an observed value of 7.71 or greater only about 1% of the time.

Methods: Tasks

- Each participant completed an individual problem-solving interview designed to elicit their reasoning about statistical models.
- The interview included a series of tasks, adapted from research by Pfannkuch, Budgett, & Arnold (2015), in which the participant compared pre- and post-differences in a set of blood pressure measurements. These measurements were taken from subjects who were randomly assigned to a fish oil diet or a regular oil diet. The goal was for each participant to decide whether they were convinced that one diet was more effective than the other in reducing blood pressure, given the data. Each participant used TinkerPlots™ to model the experimental variation to answer the question.
- Each participant was then shown a series of TinkerPlots™ samplers that either did or did not correctly model the experimental variation in the previous task. (Sampler 1, shown at right, incorrectly use replacement in both devices.) The participant was then asked to discuss whether or not each sampler could be used to provide an answer to whether one diet was more effective than the other in reducing blood pressure.



Results

- Three of the four participants correctly modeled the experimental variation in the data for using TinkerPlots™. Participant D incorrectly set up a model that assigned treatments with replacement, which resulted in an overestimation of the experimental error.
- Participants performed inconsistently when asked to identify whether or not each TinkerPlots™ sampler correctly modelled the experimental variation. (See table.) Their reasoning about why a particular model would correctly generate data to quantify the experimental variation was more informative.

Sampler	Correct decision by participant? A	B	C	D
1	Yes[1]	Yes	Yes	No
2	No	Yes	Yes	Yes
3	Yes	Yes	Yes	No
4	Yes[2]	Yes[2]	Yes	Yes
5	Yes[1]	No	No	No
6	–[3]	Yes	–[3]	No

[1] The sampler was initially incorrectly identified by the participant, but upon further discussion was correctly identified.

[2] This sampler correctly identified as not modeling the situation, but the reasons given by the participant were incorrect.

[3] Sampler 6 was not used in this case because it was identical to what the participant devised.

- Participants commented about needing to pay attention to whether a sampling device was set up *with* or *without replacement*. For example, two participants noted that sampling *with replacement* might not assign exactly half of the values to each condition. This recognition did not extend to understanding that the goal, regardless of device, was to model the experimental variation, a process that requires sampling *without replacement*. Even when participants recognized this in a model, the broad ideas seemed to be isolated to that model and not necessarily carry over to other models.
- When Participant B was asked why that a more traditional, formula-based approach could be used as an appropriate model (calculating a t-statistic), he responded by referencing the simulation-based approach: *Because that's what someone told me when I was in college, and that was, and I accepted it. …and that's why I love this part of it. That makes that question much more understandable.*

Discussion

- Teacher participants seemed to show an understanding that statistical modeling required accounting for randomness and uncertainty.
- While participants showed an understanding of some elements of statistical modeling, they were inconsistent in their own use and understanding of statistical models. Evidence from this study indicates that these teachers were able to design a model that closely mimicked the models they taught in the CATALST curriculum.
- When participants encountered models that were different from those in the curriculum, they were unable to abstract and transfer the important characteristics for modeling experimental variation.
- Participants who had taught Advanced Placement (AP) Statistics often incorrectly fused ideas of the AP methods with simulation models.
- Results suggest that mathematics teachers need additional experiences modeling and simulating to complement their preparatory work in statistics. They all had minimal prior training in statistics and even less with pedagogy related to teaching statistics. Without this development, a teacher's limited understanding or misunderstanding of statistical modeling could have detrimental effects on their students' understanding and reasoning.
- Future studies could identify types of tasks that reveal gaps in teachers' understanding and experiences in order to help develop necessary statistical content knowledge for teaching statistics.